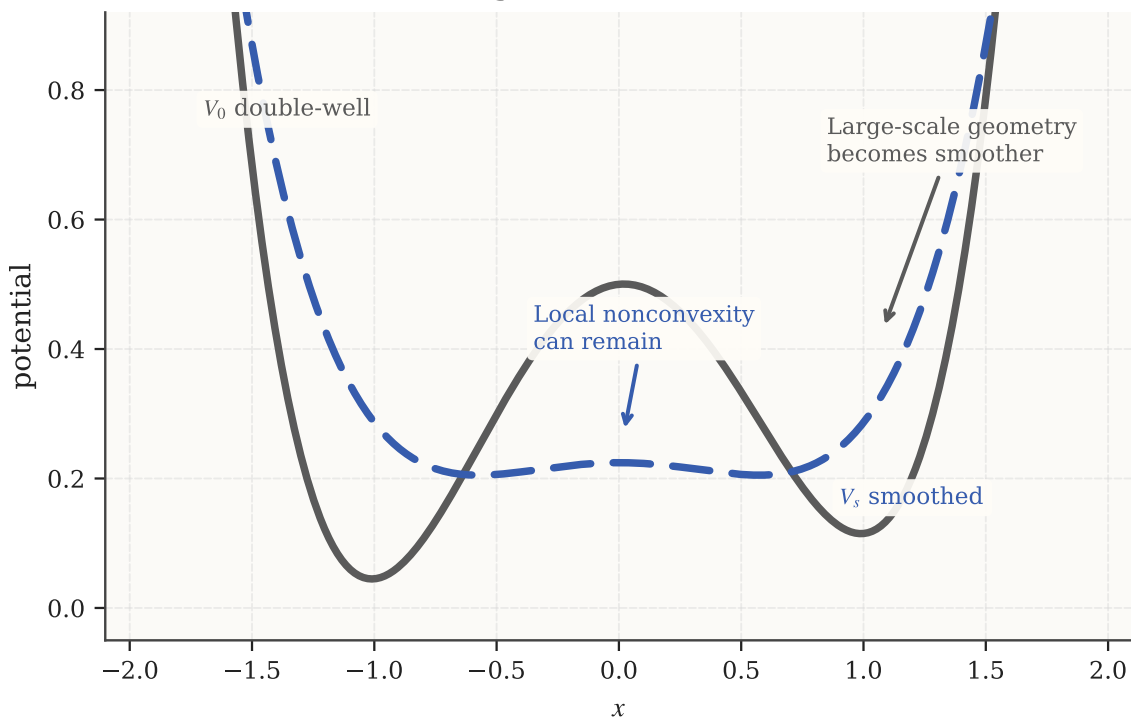
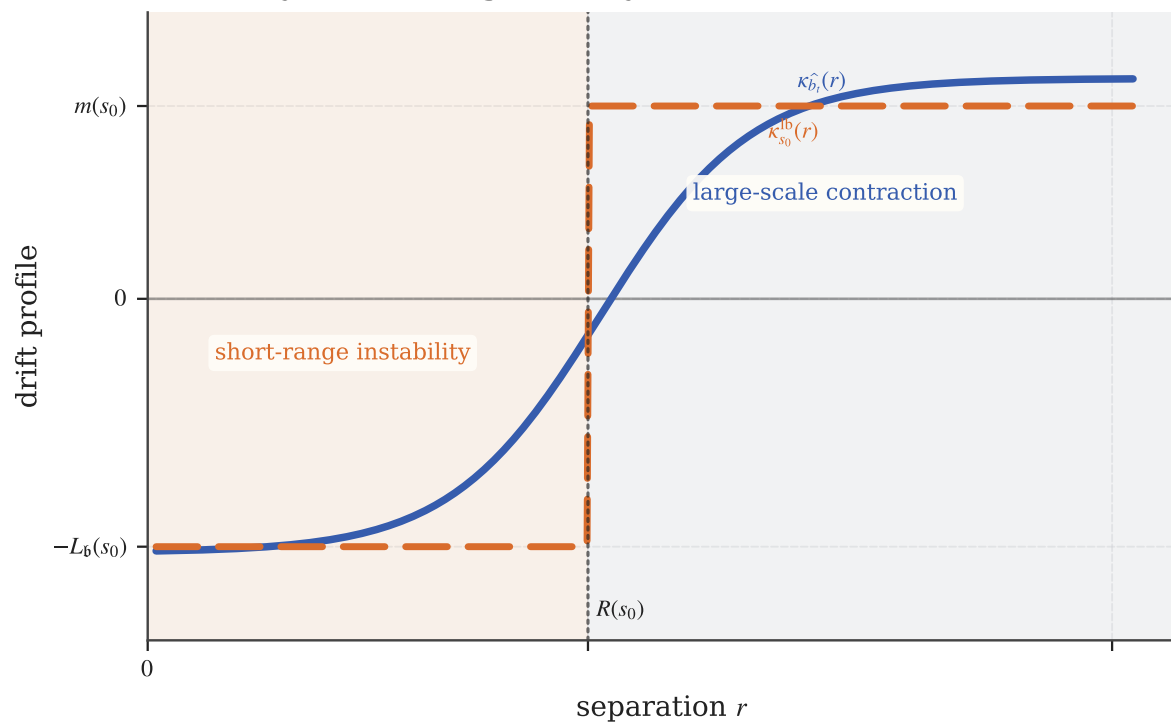


Phase-aware geometry and switching time

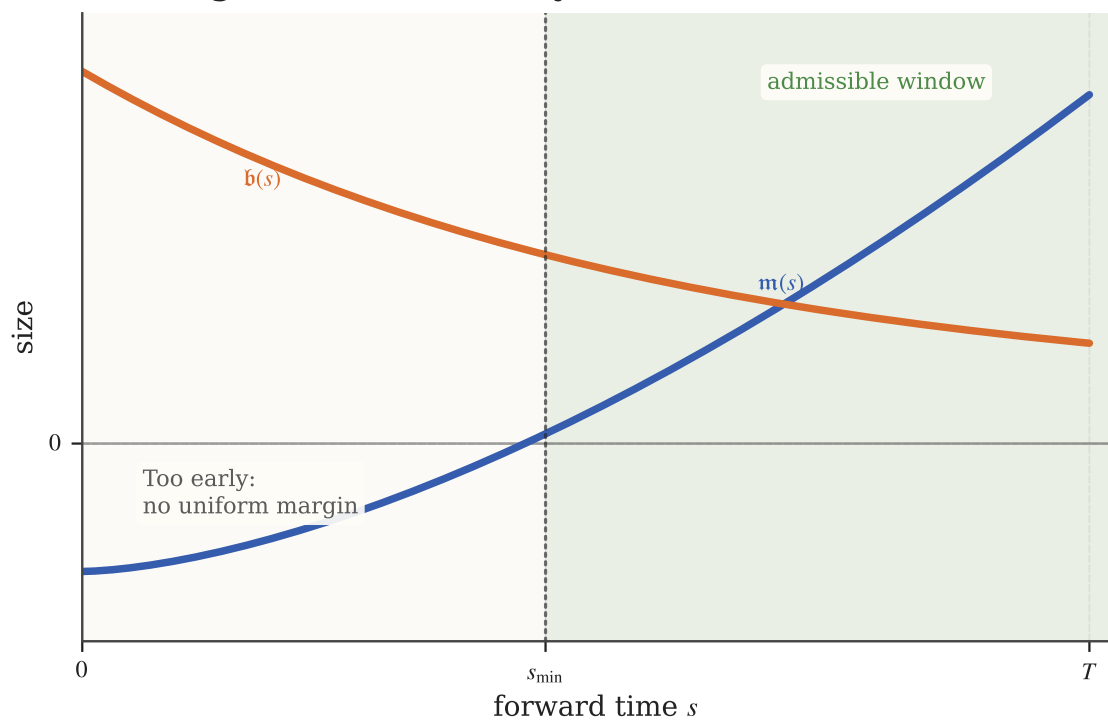
(a) Data and smoothing



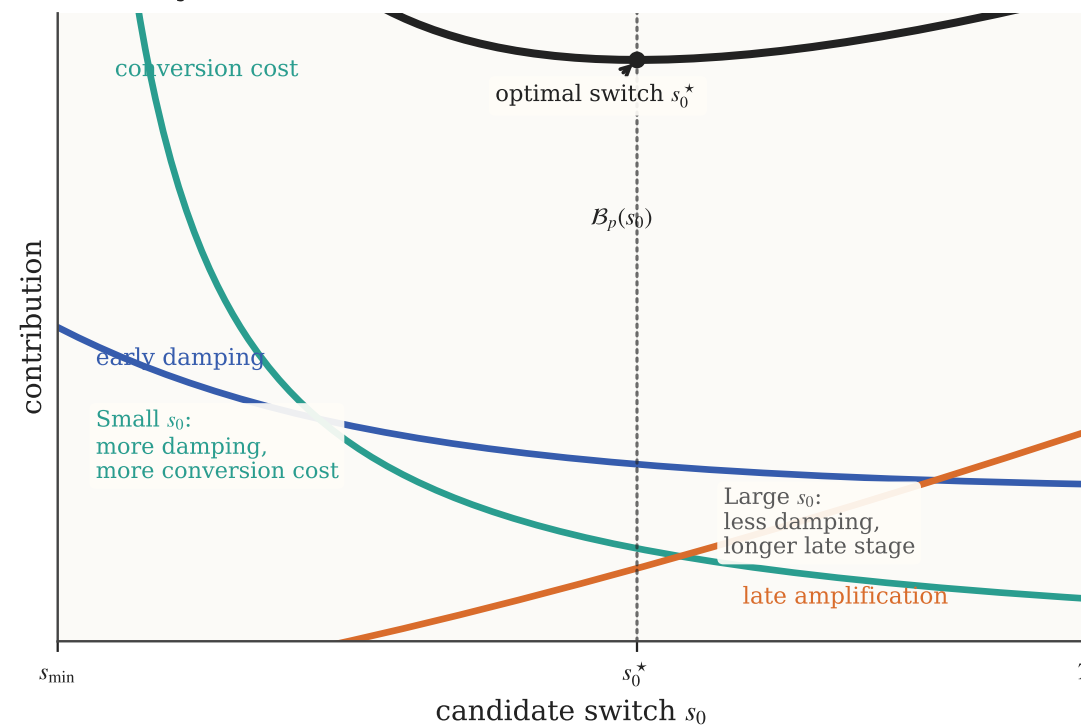
(b) Early two-scale geometry



(c) Margin and admissibility



(d) Why the switch balances



Schematic summary of the two-stage picture. Early reverse time retains short-range instability but gains large-scale contraction after smoothing; the admissible switch window appears only once that global margin is available. The preferred interior switch balances three effects: early damping, conversion cost, and late Euclidean amplification.